Computer Organization and Architecture (CSL251)
LAB-8
2025-26 W

Paged Memory Organization

Registers

- RISC-V architecture provides 31 user modifiable general-purpose registersy, x1 to x31.
- It also provides an additional read-only register x0, which is hard-wired to zero.

Register Name	ABI Name	Description
x0	zero	Hard-Wired Zero
x1	ra	Return Address
x2	sp	Stack Pointer
x3	gp	Global Pointer
x4	tp	Thread Pointer
x5	t0	Temporary/Alternate Link Register
x6-7	t1-t2	Temporary Register
x8	s0/fp	Saved Register (Frame Pointer)
x9	s1	Saved Register
x10-11	a0-a1	Function Argument/Return Value Registers
x12-17	a2-a7	Function Argument Registers
x18-27	s2-s11	Saved Registers
x28-31	t3-t6	Temporary Registers

Privilege modes

Privilege	Value	Encoding	Abbreviation
User mode	0	00	U
Machine mode	3	11	M
Supervisor mode	1	01	S

- Each mode is represented by a value and encoding stored in CSR registers.

Paged Memory Organization

- Memory management technique used by operating systems to efficiently manage memory.

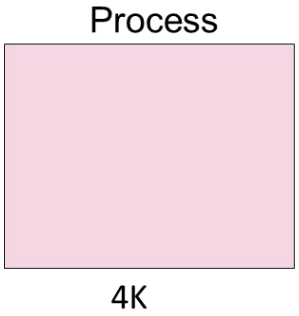
Key Idea

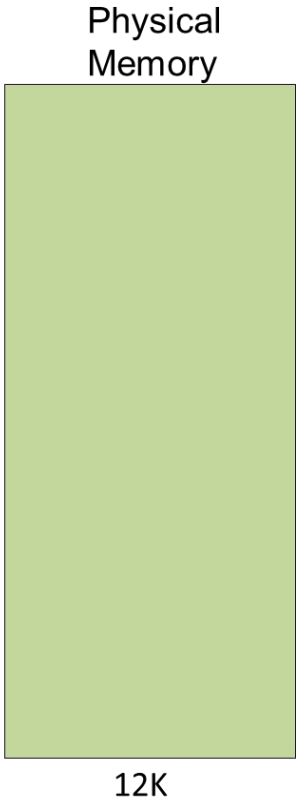
- The virtual memory of a program is divided into fixed-size blocks called pages.
- The physical memory is divided into blocks of the same size called frames.
- A page table is used to map pages to frames.

How It Works

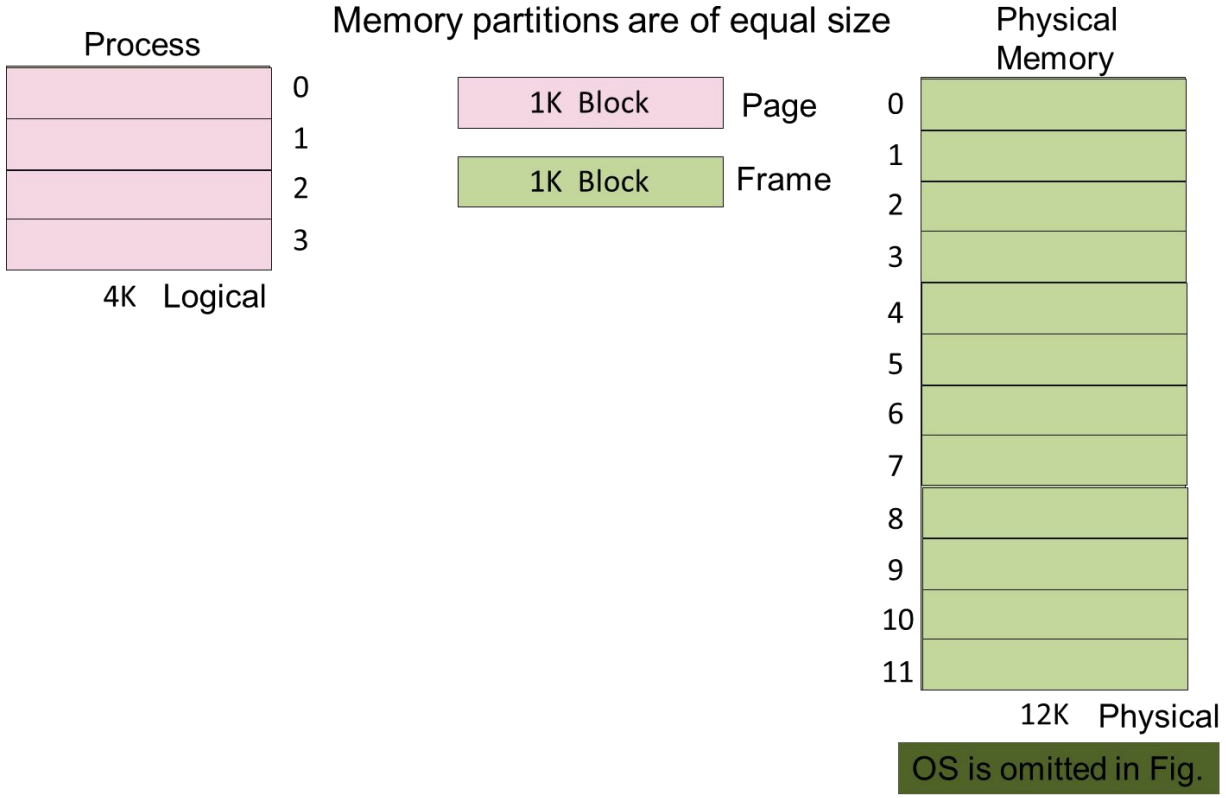
- CPU generates a virtual address.
- The page table converts it into a physical address.
- The data is then accessed from physical memory.

Paged Memory Organization

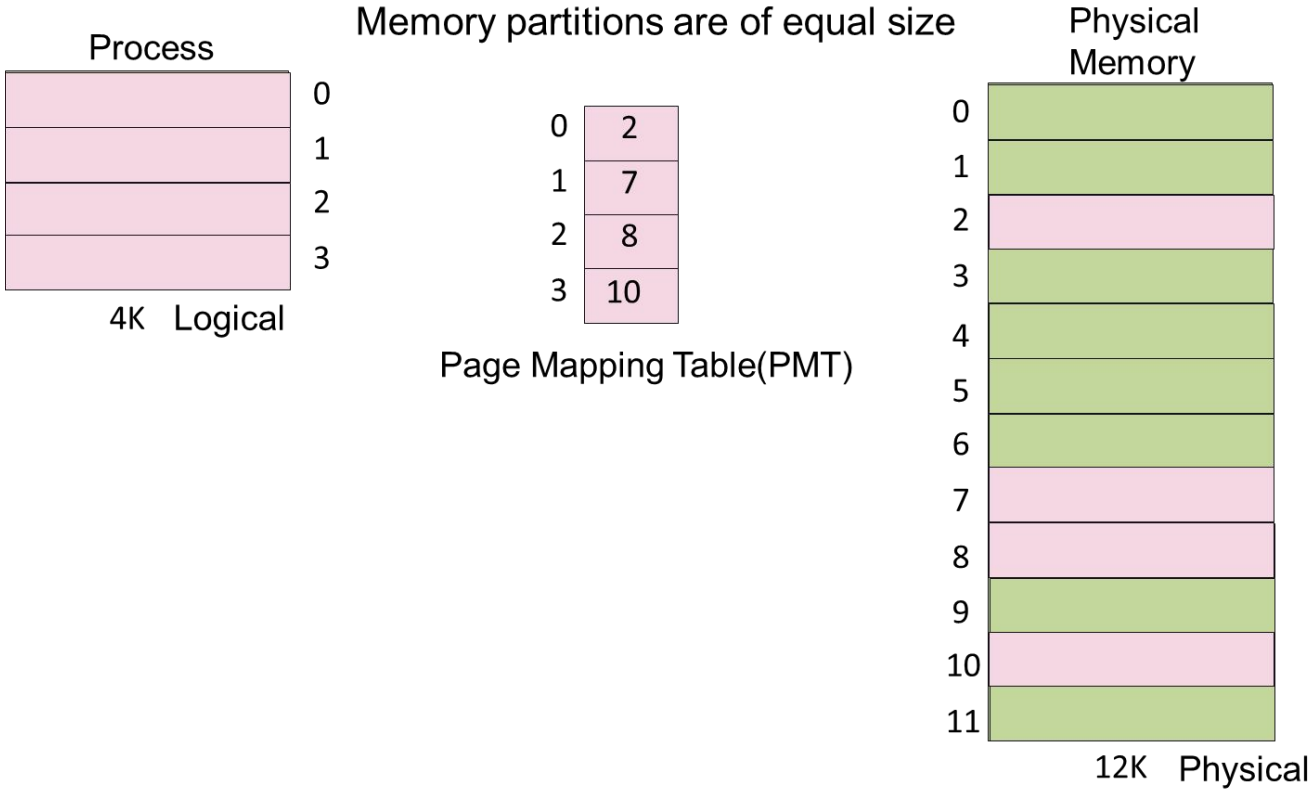




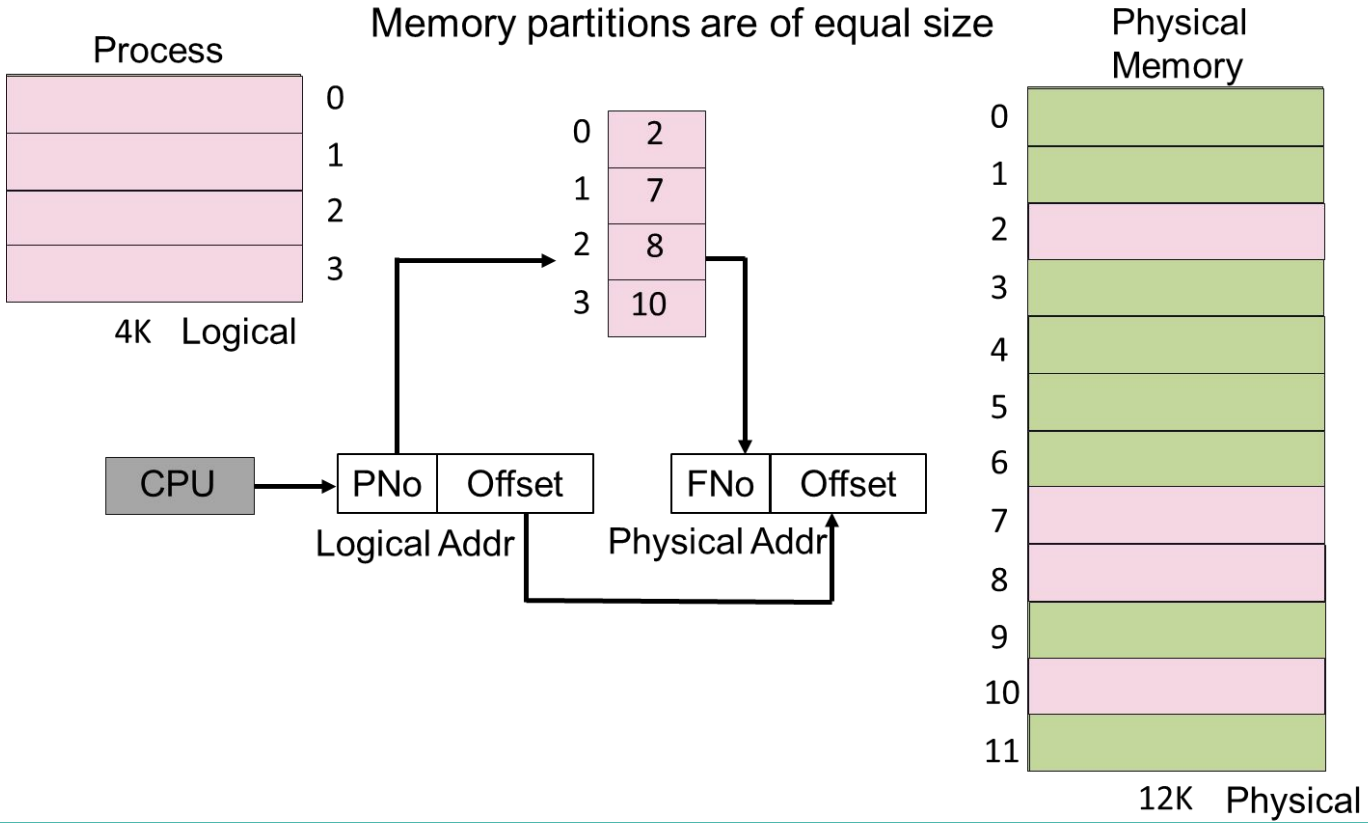
Paged Memory Organization



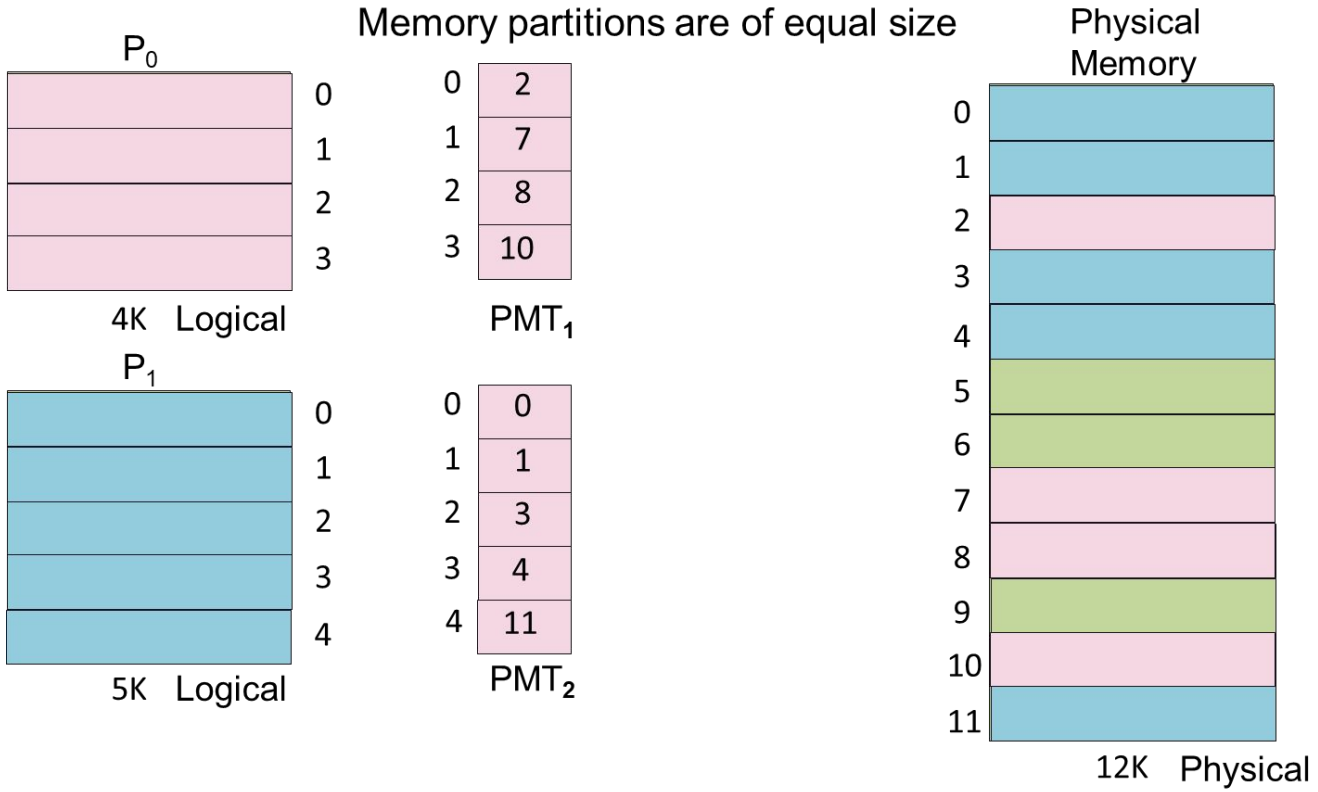
Paged Memory Organization



Paged Memory Organization



Paged Memory Organization



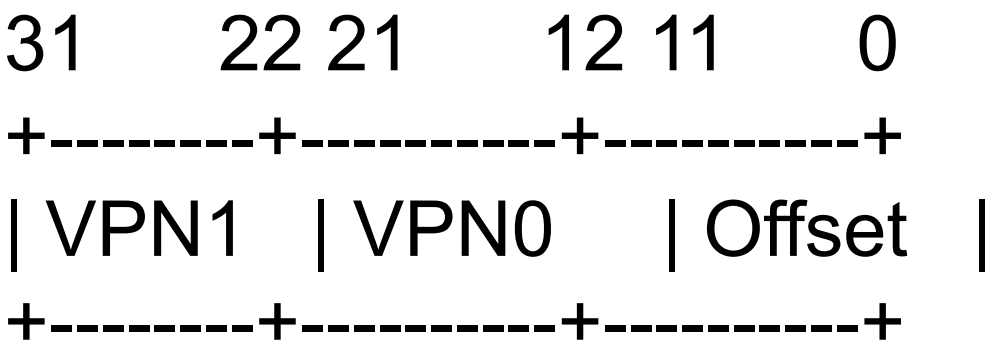
Paged Memory Organization

- Each entry in the page table contains:

Field	Meaning
PPN	Physical Page Number
V	Valid bit
R	Read permission
W	Write permission
X	Execute permission
A	Accessed bit
D	Dirty bit

Sv32 Paging in RISC-V

- 32-bit virtual address
- Two-level page table
- Page size = 4 KB



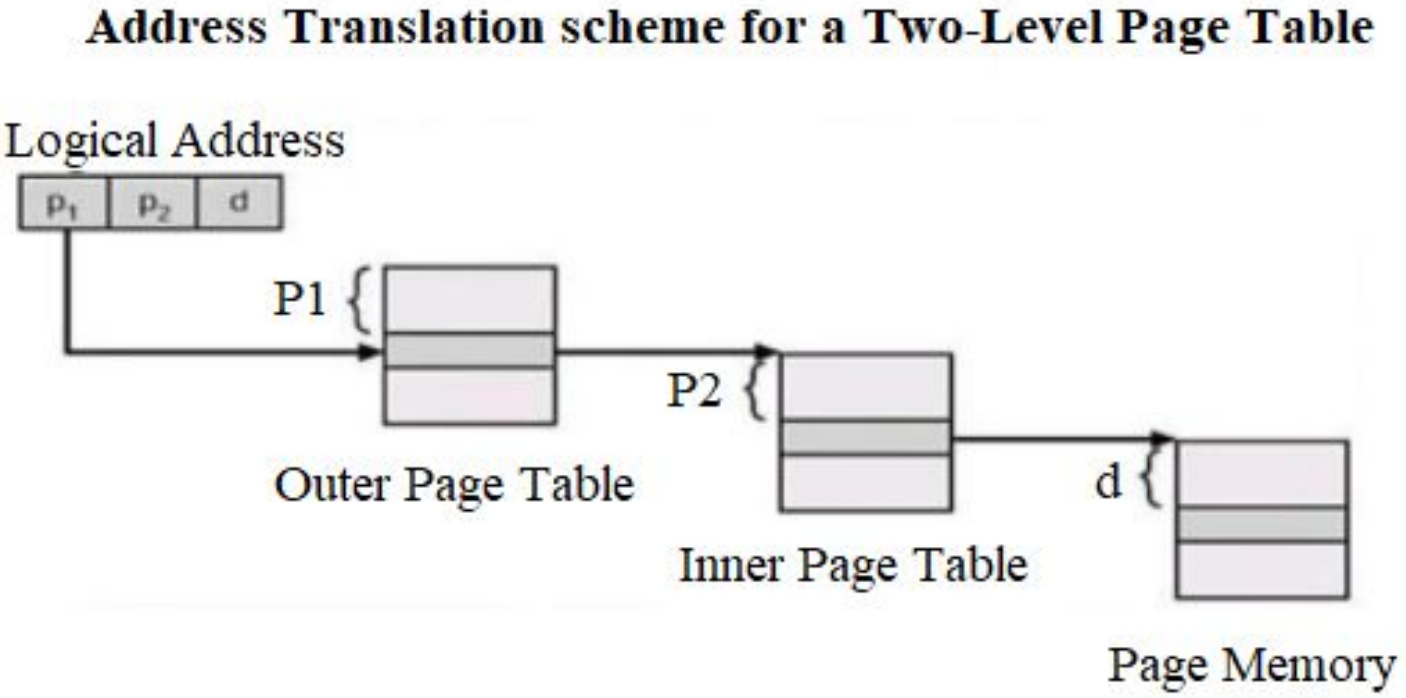
Where:

VPN1 → Level-1 page table

VPN0 → Level-0 page table

Offset → Byte inside page

Sv32 Paging in RISC-V



Two Level Paging in RISC-V

Machine Mode

↓

Supervisor Mode

↓

Create Page Table

↓

Enable Paging

↓

User Mode

↓

Execute Program

SATP(Supervisor Address Translation and Protection)

- satp is a control register in RISC-V that enables virtual memory paging.

It stores:

- **Paging mode-** Paging mode (Bare / Sv32)
- **Page table base address-** Physical Page Number of root page table

When the processor reads the satp register, it knows:

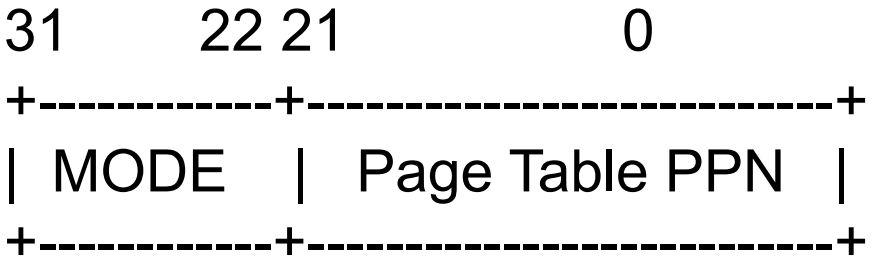
How to translate virtual addresses

and

Where the page table is located

SATP(Supervisor Address Translation and Protection)

SATP Structure (Sv32)



MODE	Meaning
0	Bare (no paging)
1	Sv32 paging

SATP(Supervisor Address Translation and Protection)

If program sets:

satp = 0x80081000

Binary breakdown:

1 00000000000010000001000000000000

↑

MODE

- The remaining bits represent the Physical Page Number (PPN) of the root page table.

PPN = 0x81000

SATP(Supervisor Address Translation and Protection)

- **Meaning:**

Paging Mode = Sv32

Root Page Table Address = 0x81000000

- **Because:**

PPN × Page Size

0x81000 × 0x1000 = 0x81000000

SATP(Supervisor Address Translation and Protection)

- **When this instruction runs:**

  csrw satp, t1

- **The processor loads:**

  satp = 0x80081000

- **Now the processor knows:**

  Paging mode → Sv32

  Root page table → 0x81000000

sfence.vma Instruction (TLB Flush)

- RISC-V instruction used to synchronize virtual memory changes.
- Clears cached address translations stored in the TLB (Translation Lookaside Buffer).
- The CPU stores recent Virtual Address → Physical Address mappings in the TLB for faster access.
- When the page table changes, the old mappings in the TLB may become incorrect.
- So we use:

sfence.vma

- to flush the old translations.

Enabling Paging

- Sfence.vma
  - Clears old address translations
- csrw satp
  - Loads page table configuration
    - satp = 0x80081000
  - Meaning:
    - MODE = Sv32
    - Root page table = 0x81000000
- Second sfence.vma
  - Ensures new translations are used

Page Table Entry Bits

- Sfence.vma
  - Clears old address translations
- csrw satp
  - Loads page table configuration
    - satp = 0x80081000
  - Meaning:
    - MODE = Sv32
    - Root page table = 0x81000000
- Second sfence.vma
  - Ensures new translations are used

Three-Level Paging

- Virtual addresses are large (39 bits in Sv39).
- Virtual Address is divided into parts:

VPN[2]	VPN[1]	VPN[0]	Page Offset
9 bits	9 bits	9 bits	12 bits

- Each level uses 9 bits to index a page table.
- Each page table contains 512 entries.

Three-Level Paging

Virtual Address

↓

Level 2 Page Table (Root)

↓

Level 1 Page Table

↓

Level 0 Page Table

↓

Physical Page